IoT Challenge #2

Packet Sniffing

Home Challenge #2

Part -1-

- Download the **two PCAP** files here or from WeBeep
- Analyse the traffic with Wireshark or writing simple scripts (Python, C++)
- Answer the 8 questions (CQ1-CQ8)

  (IGNORE MALFORMED PACKETS)

Part -2-

- Solve the exercise in **pdf** (two questions EQ1, EQ2)

FILL THE MS FORM for both parts: form

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PART 1

Questions on the PCAP file

CQ1-4 refer to file ***A.pcapng***
CQ 5-7 refer to file ***B.pcapng***
CQ8 refers to **both files**

Challenge Questions (1-2)

!! For CQ1-4, refer to the file A.pcapng

CQ1a) Find all NON Confirmable DELETE requests directed to the ***coap.me*** server that received a successful response.

Take note of the MID of such requests

CQ1b) How many of the requests found in point ***CQ1a)*** actually obtained the desired outcome?

CQ2) How many CoAP **resources** in the local server received the same number of **unique** unsuccessful Confirmable POST and PUT requests?

Assuming a resource receives **X** *different* CON unsuccessful POST requests and **Y** *different* CON unsuccessful PUT requests, how many resources have **X=Y,** with X>0?

!! Consider unsuccessful only if there is a response stating it is not successful !!

Challenge Questions (3-4)

!! For CQ1-4, refer to the file A.pcapng

CQ3a) How many separate **observe notifications** have been sent containing values for the resource "/dining_room/temperature", ignoring retransmissions?

CQ3b) How many of the notifications found in point ***CQ3a)*** are useless (e.g., a waste of traffic)?

CQ4) How many **MQTT-SN** messages are **received** by the clients from the **local broker listening on port 1885**?

!! Pay attention on the Wireshark port to analyze MQTT-SN packets

(Edit -> Preferences-> Protocols->MQTT-SN)

!! From NOW ON (CQ5-7), refer to the file B.pcapng

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Challenge Questions (5-7)

!! For CQ5-7, refer to the file B.pcapng

CQ5) How many MQTT subscribers **receive** a last will message derived from a subscription that includes atleast one wildcard?

CQ6a) How many **different MQTT clients** sent atleast one message to erase a previously retained value in a topic in the public HiveMQ broker?
CQ6b) How many of the **clients found in *CQ6a)*** connected to the HiveMQ broker with a client identifier strictly longer than 7 bytes (>7)?

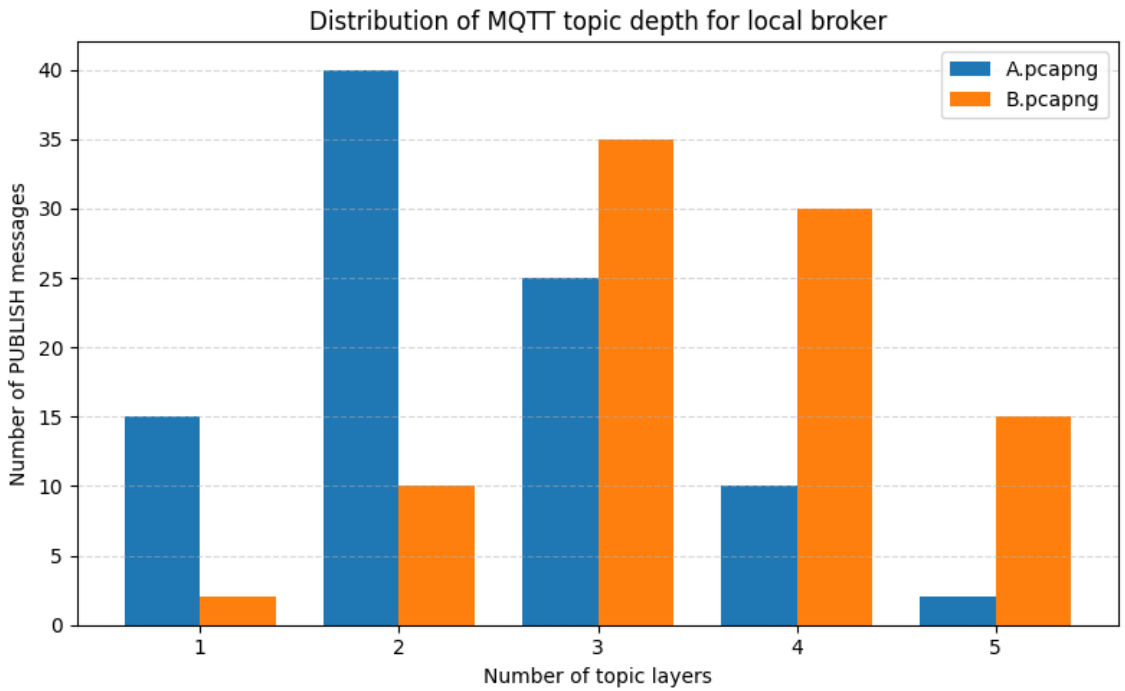
CQ7) How many **MQTT subscribe requests** directed to the **local broker** specify a topic with atleast two wildcards?

Challenge Questions (8)

CQ8) Compare the **local broker** usage in the two capture files ***A.pcapng*** and ***B.pcapng***. The goal is to have a plot on the **number of layers of the topics** used in the publish messages **directed to** the local broker in the two captures. Produce a histogram plot with number of messages for each number of layers

- Take note of the **total number of publish messages considered** for this analysis in the two captures (all number of layers) -> Report the two values as **CQ8a) CQ8b) in the form**

Produce a figure with this behaviour →



- Include **commented scripts** in the delivery (if any)
- Include the obtained **figure** in the report

PART 2

Exercise

Part 2 - Exercise

A smart building system includes:

- A **battery-powered temperature sensor** that transmits a measurement every **2 minutes**
- A **battery-powered actuator (fan controller)** that receives temperature data and decides whether to turn ON/OFF
- A **gateway (mains-powered)** that only supports MQTT

The sensor and actuator can communicate either:

- **Directly using CoAP**
- **Through the gateway using MQTT**

Communication parameters

Payload size: 12 bytes

Topic/resource length: 8 bytes

Part 2 - Exercise

Consider the following message sizes (in bytes), which **already include** header and payload size:

COAP		MQTT	
CON PUT Request	70 B	Subscribe	65 B
PUT ACK	15 B	Sub Ack	55 B
		Publish	75 B
		Pub Ack	50 B
		Connect/Disconnect	60 B
		Connect Ack	50 B

Part 2 - Exercise

Assume that:

1. Transmit and Receive cost per bit are
   E_{TX} = 50nJ/bit, E_{RX} = 58nJ/bit (nanojoule per bit)
2. CoAP communication uses **Confirmable (CON)** messages
3. MQTT communication uses **QoS 1** (for both actuator and sensor sides)
4. Ideal communication (no losses)
5. When retransmission is required, *RTO = 30s and MAX_TRANSMISSIONS = 3*
6. Devices start disconnected
7. MQTT connections are closed gracefully during sleep periods (with non-acknowledged DISCONNECT)
8. Only the energy consumed by the **sensor and actuator** must be considered (ignore gateway consumptions)

Part 2 - Exercise

Exercise Question 1 (EQ1):

Compute the **total communication energy consumed by the battery-powered sensor and actuator over 1 hour** in:
(a) CoAP (direct communication)
(b) MQTT (via the gateway)
Provide numerical results in **µJ (microjoule)**.

Exercise Question 2 (EQ2):

Assume now that the **actuator wakes up** only once every 30 minutes.
Which protocol would you choose for long-term operation?
Justify your answer (max 300 characters), explicitly stating any protocol parameter required (e.g., clean session, retain, observe, block-transfer modes etc…) and performance metric you optimize.

For **EQ1a and EQ1b**, motivate your answers in the **PDF**, and report the results in the **form.**
For **EQ2, report your full response in both the report and the form.**

Challenge deliverable

Deliver a ZIP file containing:

- A **PDF** named **Challenge.pdf** containing **motivated responses** with all filters used and comments of any additional code (if used). Specify your **names + person codes** in the pdf!
- A **PDF** named **Exercise.pdf** containing the solution of the Part 2 exercise. Specify your **names + person codes** in the pdf!
- **FILL this form with the requested values (both parts)**

Organize clean reports! Very bad reports will be penalized

Upload the zip file on the **folder #2** on WeBeep "Consegne" folder

The ZIP should be named as follows:

2-teams: <leader_personcode>_<other_personcode>.zip

Single: <person_code>.zip

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PLEASE NOTE (Part 1)

1. Files without name and personcode or in a weird format won't be corrected (=0 points)

2. Always make clear and **highlight** your answer.
   The answer **should be clear and unique!**
   Example, answers like this:
   "If …… then the answer is 4, but if instead …… then answer is 5"
   Will get 0 points. **If you have doubts on questions ask before submitting**

3. Try to produce **small but effective answers**, write only useful things and if required upload images taken e.g. from wireshark. Final answers should be written as text. (not like, "see image for the answer")

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For 2 people teams:

- Choose your team leader and name the file as:
  <leader_personcode>_<other_personcode>.zip

- **Only the teamleader** should upload the challenge in WeBeep
  Do note upload the same challenge more times!!!

- *Can I take the challenges with the other class students (different Prof.)*?
  YES, but **only the team leader** should upload the challenge in WeBeep

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Delivery Deadline

- **FIRM** Deadline:

  April 23, 2026 h 23.59

- Max 2 people

Good Luck!

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